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Municipality-level heterogeneity in electricity-demand responses to temperature: Functional patterns, spatial structure, and regional correlates in South Korea

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Abstract:	Temperature-sensitive electricity demand can vary substantially across locations, yet the functional and spatial organization of these heterogeneous responses remains poorly understood. This study examines municipality-level temperature response functions (TRFs) for residential and general-service electricity demand in South Korea. Monthly electricity consumption was linked to the full distribution of hourly temperatures, and separate TRFs were estimated for each municipality–contract combination. Functional principal component analysis (FPCA), k-means clustering, spatial autocorrelation analysis, and spatial error models were then used to characterize cross-municipality heterogeneity and its regional correlates. U-shaped responses predominated, accounting for 96.8% of residential and 90.3% of general-service TRFs. Residential demand exhibited substantially stronger heat-side than cold-side responses, whereas general-service responses were more balanced. Three functional principal components explained 95.8% and 95.3% of cross-municipality variation in residential and general-service TRF shapes, respectively, while clustering of the component scores identified distinct recurring response profiles. TRF characteristics also exhibited significant positive spatial autocorrelation, particularly for residential demand. Spatial error models further showed that warmer municipalities tended to have higher optimal temperatures in both contract types. Residential outer heat-side sensitivity was stronger in municipalities with lower seasonal temperature variability, higher population density, and higher real GRDP per capita, while the correlates of general-service responses differed. These findings demonstrate that electricity-demand responses to temperature are functionally and geographically structured rather than spatially homogeneous, highlighting the value of municipality-specific TRFs for regional demand forecasting, peak-load management, and climate-adaptation planning.

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Dear Editor,

I am pleased to submit the manuscript entitled **"Municipality-level heterogeneity in electricity-demand responses to temperature: Functional patterns, spatial structure, and regional correlates in South Korea"** for consideration for publication in *Energy*.

Understanding how electricity demand responds to temperature across space is increasingly important for energy-demand forecasting, peak-load management, and climate adaptation. Although previous studies have documented nonlinear and geographically heterogeneous temperature–electricity relationships, considerably less attention has been given to whether the response functions themselves exhibit systematic functional and spatial structure across a large number of local areas.

This study addresses this issue using municipality-level electricity consumption and hourly meteorological observations in South Korea from January 2004 to May 2026. Separate temperature response functions (TRFs) are estimated for residential and general-service electricity demand across 217 municipalities using a distribution-based framework that preserves the full within-month distribution of hourly temperature exposure. The estimated TRFs are subsequently analyzed using functional principal component analysis, data-driven clustering, spatial autocorrelation analysis, and spatial error models.

The study provides three main contributions. First, it demonstrates substantial municipality-level heterogeneity in electricity-demand responses to temperature despite the predominance of U-shaped relationships. Second, by treating the estimated TRFs as functional observations, it shows that cross-municipality differences in complete response shapes can be represented by a small number of dominant functional dimensions and recurring response profiles. Third, it demonstrates that this functional heterogeneity is geographically structured and systematically associated with regional climatic, demographic, economic, and built-environment characteristics. The results further reveal important differences between residential and general-service demand, with residential electricity demand exhibiting particularly strong heat-side responses and more pronounced spatial organization.

These findings are directly relevant to the scope of *Energy* because they demonstrate how geographically differentiated temperature sensitivity can affect the representation and management of electricity demand under changing climatic conditions. The proposed framework provides a means of moving beyond nationally aggregated

temperature–demand relationships toward spatially explicit representations that can support regional demand forecasting, peak-load management, and energy-system planning.

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Sincerely,

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Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics declaration

The author(s) declare(s) that the study does not involve humans nor animal subjects.

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Municipality-level heterogeneity in electricity-demand responses to temperature: Functional patterns, spatial structure, and regional correlates in South Korea

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Abstract

Temperature-sensitive electricity demand can vary substantially across locations, yet the functional and spatial organization of these heterogeneous responses remains poorly understood. This study examines municipality-level temperature response functions (TRFs) for residential and general-service electricity demand in South Korea. Monthly electricity consumption was linked to the full distribution of hourly temperatures, and separate TRFs were estimated for each municipality–contract combination. Functional principal component analysis (FPCA), k-means clustering, spatial autocorrelation analysis, and spatial error models were then used to characterize cross-municipality heterogeneity and its regional correlates. U-shaped responses predominated, accounting for 96.8% of residential and 90.3% of general-service TRFs. Residential demand exhibited substantially stronger heat-side than cold-side responses, whereas general-service responses were more balanced. Three functional principal components explained 95.8% and 95.3% of cross-municipality variation in residential and general-service TRF shapes, respectively, while clustering of the component scores identified distinct recurring response profiles. TRF characteristics also exhibited significant positive spatial autocorrelation, particularly for residential demand. Spatial error models further showed that warmer municipalities tended to have higher optimal temperatures in both contract types. Residential outer heat-side sensitivity was stronger in municipalities with lower seasonal temperature variability, higher population density, and higher real GRDP per capita, while the correlates of general-service responses differed. These findings demonstrate that electricity-demand responses to temperature are functionally and geographically structured rather than spatially homogeneous, highlighting the value of municipality-specific TRFs for regional demand forecasting, peak-load management, and climate-adaptation planning.

Keywords: Electricity demand; Temperature response; Climate change; Spatial heterogeneity; Functional data analysis

1. Introduction

Climate change is reshaping electricity demand by altering the frequency, intensity, and duration of temperature extremes [1]. Electricity consumption is particularly sensitive to ambient temperature because heating and cooling requirements increase as temperatures deviate from comfortable conditions [2]. Consequently, the temperature–electricity relationship is typically nonlinear [3–5]. Recent studies have increasingly employed flexible temperature response functions (TRFs), allowing the shape of this relationship to emerge from the data rather than imposing fixed heating or cooling thresholds [6,7].

As evidence of nonlinearity has accumulated, attention has increasingly shifted toward heterogeneity in temperature sensitivity across locations and consumer groups [8]. Temperature sensitivity reflects not only climatic exposure but also differences in air-conditioning and heating technologies, building characteristics, income, economic structure, and patterns of electricity use [2,9–11]. Accordingly, similar temperature changes need not generate similar electricity-demand responses across regions [12]. Empirical evidence has documented substantial spatial heterogeneity in temperature-sensitive electricity demand across countries [13], cities [14], subnational regions [15], and provincial power systems [16]. These studies show that temperature responses vary not only across locations but also across consumer groups and load structures.

This spatial heterogeneity has implications beyond the estimation of electricity demand itself. More broadly, climate change can affect electricity systems through both demand-side changes in heating and cooling requirements and supply-side impacts on electricity generation [1]. If regions respond differently to the same climatic changes, future warming can alter not only aggregate electricity consumption but also the geographic distribution and temporal coincidence of electricity loads. Spatial heterogeneity in temperature responses also has implications for power-system planning. Geographically uneven changes in weather-sensitive loads can affect peak demand, capacity requirements, transmission needs, storage deployment, and interregional power flows [13,17–19]. These effects are particularly relevant where electricity supply and demand are geographically separated.

South Korea provides a particularly relevant setting in which to examine this issue. Despite its relatively small geographical size, Korea exhibits substantial climatic variation, with colder winter conditions in northern and mountainous areas and milder conditions in southern and coastal regions. Although annual peak demand was historically associated primarily with summer cooling, winter electricity demand became increasingly important as electricity use for heating expanded, particularly in the commercial sector. Previous Korean research has found that heating-related electricity demand grew faster than baseline and cooling demand over 1999–2016, with particularly pronounced growth in the commercial sector [20]. Temperature-sensitive load in Korea therefore cannot be understood solely as a summer cooling phenomenon.

The spatial dimension is also important in the Korean power system because electricity production and consumption are geographically mismatched, with large generating facilities concentrated in a limited number of municipalities and major demand centers located elsewhere [21,22]. This separation increases the relevance of understanding not only the aggregate magnitude but also the geographic distribution of weather-sensitive electricity demand.

Research on temperature-sensitive electricity demand in Korea has progressively developed toward flexible representations of nonlinear and extreme-temperature responses. Early work documented substantial sectoral heterogeneity [23]. Chang et al. [24] introduced a distribution-based framework linking monthly electricity demand to the full density of high-frequency temperature observations, allowing nonlinear responses to be estimated without reducing within-month weather conditions to a single statistic. Subsequent studies applied this framework to heating and cooling demand [20], extended it to peak demand and additional weather conditions [25], and examined responses in the tails of the temperature distribution [26]. Collectively, this literature

demonstrates nonlinear, asymmetric, and sector-specific temperature responses in Korea.

Yet a different dimension of heterogeneity remains less understood. Existing studies have primarily characterized temperature responses for aggregate sectors, selected consumer groups, individual cities or provinces, or predefined geographic units. Although substantial regional heterogeneity in temperature responses has been documented across cities, provinces, and other geographic units [13–16], less attention has been given to the cross-sectional variation of the response functions themselves as functional observations.

This leaves open a broader question: when temperature response functions are estimated for a large number of locations, does the heterogeneity among the functions themselves exhibit systematic structure? In particular, it remains unclear whether cross-municipality differences in complete TRF shapes can be represented by a small number of dominant functional dimensions, whether recurring response profiles can be identified without imposing predefined regional categories, and whether these functional differences are geographically structured.

Functional data analysis provides a natural framework for addressing these questions because it treats an entire curve, rather than a set of isolated coefficients, as the unit of observation [27]. Clustering and dimensionality-reduction approaches have been widely applied to electricity load profiles [28]. For example, Seo et al. [29] treated commercial electricity-consumption curves as functional observations and used functional principal component analysis and clustering to identify systematic variation in usage patterns. This illustrates the usefulness of a functional perspective for organizing large collections of electricity-related curves. However, its application to the estimated temperature response functions themselves, particularly for characterizing cross-regional heterogeneity and its spatial organization, has received comparatively limited attention.

Against this background, this study investigates the functional and spatial heterogeneity of temperature-sensitive electricity demand across Korean municipalities. Municipality-specific TRFs are estimated separately for residential and general-service demand using monthly electricity consumption and distributions of hourly temperatures. The estimated functions are then analyzed using FPCA, data-driven clustering, spatial autocorrelation analysis, and spatial error models.

This study contributes to the literature in three respects. First, it provides nationwide municipality-specific TRFs for residential and general-service electricity demand rather than imposing a common national or broad regional response. Second, it preserves the full distribution of hourly temperature exposure within each month. Third, and most importantly, it shifts the analysis of regional heterogeneity from individual response indicators to the complete shape of the TRF by combining FPCA, functional clustering, and spatial analysis.

The remainder of this paper is organized as follows. Section 2 describes the electricity and meteorological data, the estimation of municipality-specific temperature response functions, and the functional and spatial analysis methods. Section 3 presents the empirical results, including municipality-specific TRFs, functional variation and clustering, spatial patterns, and regional correlates. Section 4 summarizes the main findings and discusses their implications for electricity-demand forecasting, regional load management, and climate adaptation.

2. Data and Methodology

2.1 Data

Monthly electricity consumption data covering January 2004 to May 2026 were obtained from the Korea Electric Power Corporation (KEPCO). The dataset reports municipality-level electricity consumption separately for residential and general-service contract types. To ensure consistent spatial identification across electricity, temperature, and geographic datasets, all observations were harmonized to a common reference geography comprising 229 municipalities in South Korea.

Hourly air temperature observations for the same period were obtained from the Automatic Weather System (AWS) operated by the Korea Meteorological Administration (KMA). The AWS network comprised 745 monitoring stations distributed across South Korea. Each station was assigned to a municipality based on its administrative location. When multiple AWS stations were available within the same municipality, municipality-level hourly temperature was calculated as the arithmetic mean of all available station observations.

Following spatial harmonization and the integration of electricity and temperature records, 218 municipalities contained sufficient matched observations to construct the regression dataset. Municipality–contract combinations with fewer than 60 monthly observations were excluded from TRF estimation to avoid estimating long-run temperature-response relationships from short time series. This criterion excluded one additional municipality, leaving 217 municipalities for each contract type and yielding a total of 434 municipality-specific TRFs.

Municipality-level hourly temperature observations were converted into monthly temperature distributions using kernel density estimation (KDE). Rather than representing monthly weather conditions by a single mean temperature, this approach preserves information on the full distribution of temperatures experienced within each month. The estimated monthly temperature densities (p_{it}) were evaluated over a common temperature domain and subsequently integrated with the Fourier basis functions used in the TRF specification. This procedure generated the four temperature-distribution variables (x_1–x_4) defined in Eqs. (6)-(9), which were then merged with monthly electricity consumption to construct the final regression dataset.

2.2 Temperature response function

To reduce the influence of long-term trends and seasonal changes in consumption levels, monthly electricity consumption was normalized by its 12-month moving average. The dependent variable was defined as:

$$Y_{ict} = \ln\left(\frac{E_{ict}}{\overline{E}_{ict}^{(12)}}\right) \quad (1)$$

where E_{ict} denotes monthly electricity consumption in municipality i, contract type c, and month t, and $\overline{E}_{ict}^{(12)}$ represents the corresponding 12-month moving average. Thus, Y_{ict} represents the proportional deviation of monthly electricity consumption from its 12-month moving-average level.

Following the distribution-based TRF framework [24], electricity demand was modeled as the expected response over the full distribution of temperatures experienced during each month:

$$Y_{ict} = \alpha_{ic} + \int_{T_{min}}^{T_{max}} g_{ic}(T)p_{it}(T)dT + \varepsilon_{ict} \quad (2)$$

where T denotes air temperature, $g_{ic}(T)$ denotes the municipality- and contract-specific temperature response function, $p_{it}(T)$ is the monthly temperature density estimated from hourly observations using kernel density estimation, α_{ic} is an intercept specific to each municipality–contract combination, and ε_{ict} is the error term. This formulation accounts for the entire within-month temperature distribution rather than representing monthly weather conditions using a single temperature statistic.

The temperature response function is approximated using a second-order Fourier Flexible Form (FFF), which allows nonlinear temperature responses to be represented using a parsimonious set of basis functions:

$$g_{ic}(T) = \beta_{1,ic}s(T) + \beta_{2,ic}s(T)^2 + \beta_{3,ic}cos(2\pi s(T)) + \beta_{4,ic}sin(2\pi s(T)) \quad (3)$$

where temperature was rescaled to the unit interval as

$$s(T) = \frac{T - T_{min}}{T_{max} - T_{min}} \quad (4)$$

In the present analysis, the TRF was evaluated on a common temperature grid from -30℃ (T_{min})to 45℃ (T_{max}) at 0.1℃ intervals. The common grid was used to ensure that municipality-specific functions were evaluated on an identical temperature scale.

Substituting Eq. (3) into Eq. (2) yields a linear regression model because each basis function enters the monthly demand equation through its expectation over the monthly temperature distribution:

$$Y_{ict} = \alpha_{ic} + \sum_{k=1}^{4} \beta_{k,ic}x_{k,it} + \varepsilon_{ict} \quad (5)$$

where

$$x_{1,it} = \int s(T)p_{it}(T)dT \quad (6)$$

$$x_{2,it} = \int s(T)^2 p_{it}(T)dT \quad (7)$$

$$x_{3,it} = \int cos(2\pi s(T))p_{it}(T)dT \quad (8)$$

$$x_{4,it} = \int sin(2\pi s(T))p_{it}(T)dT \quad (9)$$

The coefficients $\beta_{k,ic}$ were estimated separately for each municipality and contract type using ordinary least squares, producing a time-invariant TRF for each municipality–contract combination.

To characterize the estimated TRFs while limiting the influence of sparsely observed temperature tails, curve shape and associated response characteristics were evaluated over a common interpretation range of −10℃ to 32℃. The full −30℃ to 45℃ domain was retained for TRF evaluation, whereas the narrower interval was used only for subsequent curve characterization. A TRF was classified as U-shaped when it exhibited an interior minimum within this range and the responses at both endpoints (-10℃ and 32℃) exceeded the minimum by at least 5% of the curve amplitude.

For U-shaped TRFs, the optimal temperature was defined as the temperature at which the estimated response reached its minimum over −10 to 32 ℃:

$$T_{ic}^{*} = \arg\min_{T\in[-10,32]} \hat{g}_{ic}(T) \quad (10)$$

Local cold- and heat-side sensitivities were calculated over 5 ℃ intervals around the optimum. To characterize responses farther from the optimum, outer cold- and heat-side sensitivities were calculated using common endpoints of −8 ℃ and 32 ℃, approximately corresponding to the median municipality-level 1st and 99th percentile temperatures. Detailed definitions are provided in Supplementary Methods.

2.3 Functional principal component analysis

To characterize cross-municipality variation in the shapes of the estimated TRFs, functional principal component analysis (FPCA) [27] was applied separately to residential and general-service electricity demand. The analysis was restricted to TRFs classified as U-shaped according to the criteria described in Section 2.2. Because the objective was to compare differences in response shape rather than differences in response magnitude, each TRF was first normalized to the unit interval:

$$\tilde{g}_{ic}(T) = \frac{\hat{g}_{ic}(T) - \min_T \hat{g}_{ic}(T)}{\max_T \hat{g}_{ic}(T) - \min_T \hat{g}_{ic}(T)} \quad (11)$$

where $\hat{g}_{ic}(T)$ denotes the estimated TRF for municipality i and contract type c. This transformation sets the minimum and maximum of each response function to zero and one, respectively, thereby removing differences in overall amplitude while preserving differences in curve shape.

For functional comparison, the normalized TRFs were evaluated on a common temperature grid spanning −10℃ to 32℃ . This range corresponds to the interpretation domain used for TRF characterization in Section 2.2 and limits the influence of sparsely observed temperature tails on the functional decomposition. The normalized function for each municipality can be represented as

$$\tilde{g}_{ic}(T) = \mu_c(T) + \sum_{k=1}^{K_c} \xi_{ikc}\phi_{kc}(T) + \eta_{ic}(T) \quad (12)$$

where $\mu_c(T)$ denotes the mean normalized TRF for contract type c, $\phi_{kc}(T)$ is the k-th functional principal component, ξ_{ikc} is the corresponding principal component score for municipality i, and $\eta_{ic}(T)$ represents the residual variation not captured by the retained components.

PCA was applied to the municipality-by-temperature matrix of normalized TRFs, providing a discretized approximation to FPCA. Components explaining at least 90% of cumulative variance were retained for subsequent clustering.

2.4 Functional clustering and spatial analysis

To identify representative patterns of municipality-specific temperature responses, clustering analysis was performed separately for residential and general-service electricity demand using the FPCA scores retained in Section 2.3. Because the retained scores provide a low-dimensional representation of the normalized TRF shapes, municipalities with similar score vectors represent similar temperature-response profiles.

Municipalities were grouped using the k-means algorithm [30], which partitions the FPCA score vectors into K clusters by minimizing the within-cluster sum of squared Euclidean distances:

$$\min_{\{C_k\}_{k=1}^{K}} \sum_{k=1}^{K} \sum_{i\in C_k} \|\boldsymbol{\xi}_{ic} - \boldsymbol{\mu}_{kc}\|^2 \quad (13)$$

where $\boldsymbol{\xi}_{ic} = (\xi_{i1c}, \xi_{i2c}, \cdots, \xi_{iK_c c})^{\top}$ denotes the vector of retained FPCA scores for municipality i for contract type c, C_k denotes the set of municipalities assigned to cluster k, and $\boldsymbol{\mu}_{kc}$ is the centroid of cluster k in the retained FPCA-score space. Multiple random initializations were used to reduce sensitivity to the initial cluster centers.

Candidate solutions with K= 2–6 were evaluated using the silhouette coefficient, Calinski–Harabasz index, and gap statistic [31–33]. Because these criteria did not uniformly select the same K across contract types, a common three-cluster solution was adopted considering diagnostic performance, cluster size, interpretability, and cross-contract comparability. Robustness to random initialization was assessed using the adjusted Rand index [34]. Cluster profiles were represented by pointwise mean normalized TRFs with 95% confidence intervals obtained by bootstrap resampling [35].

Finally, the resulting clusters were characterized using the TRF attributes defined in Section 2.2, including optimal temperature, local and outer cold- and heat-side sensitivities, and curve amplitude. Regional climatic characteristics, including mean temperature and seasonal temperature variability, were also compared across clusters. Differences in regional climatic characteristics among clusters were evaluated using the Kruskal–Wallis test [36]. The spatial distribution of cluster membership was subsequently mapped to examine geographic patterns in municipality-level temperature responses.

Spatial autocorrelation was assessed separately for residential and general-service demand using six continuous TRF characteristics: optimal temperature, local and outer cold- and heat-side sensitivities, and curve amplitude. Global Moran's I [37] was calculated as:

$$I = \frac{n}{W}\frac{\sum_i \sum_j w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \quad (14)$$

where n is the number of municipalities, x_i denotes the TRF characteristic for municipality i, $\bar{x}$ is the mean of that characteristic across municipalities, w_{ij} is the spatial weight between municipalities i and j, and $W = \sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}$ is the sum of all spatial weights. Spatial weights were based on symmetrized, row-standardized four-nearest-neighbor relationships between municipality representative points. Statistical significance was assessed using 999 permutations [38].

2.5 Regional correlates of TRF heterogeneity

To examine whether cross-municipality variation in temperature responses was systematically associated with local characteristics, we related selected TRF characteristics to climatic, demographic, economic, and built-environment variables. Three outcomes were considered: the optimal temperature (T_{ic}^{*}), outer cold-side sensitivity (S_{ic}^{OC}), and outer heat-side sensitivity (S_{ic}^{OH}). These measures were selected to represent the location of the TRF minimum and the responses toward the colder and hotter portions of the temperature range, respectively.

Municipality characteristics included mean temperature, seasonal temperature variability, aging rate, population density, real GRDP per capita, and apartment share. Mean temperature and seasonal temperature variability characterize the local climate, while the remaining variables capture demographic, urban, economic, and residential characteristics that may be associated with differences in temperature-sensitive electricity demand. These variables were selected based on previous evidence that temperature-sensitive energy demand varies with climatic conditions, demographic composition, income and economic conditions, urbanization, and characteristics of the residential building stock [9,15,39–41]. Population density and real GRDP per capita were log-transformed because of their skewed distributions. All explanatory variables were standardized prior to estimation, allowing the estimated coefficients to represent the change in each TRF characteristic associated with a one-standard-deviation change in the corresponding municipality characteristic. We first estimated cross-sectional ordinary least squares (OLS) models of the form:

$$Y_{ic} = a_c + \boldsymbol{X}_i'\boldsymbol{\beta}_c + \varepsilon_{ic} \quad (15)$$

where Y_{ic} denotes one of the three TRF characteristics for municipality i for contract type c, and $\boldsymbol{X}_i$ is a vector of municipality characteristics. Two alternative specifications were estimated because aging rate and apartment share were highly negatively correlated (r = -0.92). Model A included mean temperature, seasonal temperature variability, aging rate, log population density, and log real GRDP per capita, whereas Model B replaced aging rate with apartment share. This approach avoids including the two strongly correlated variables simultaneously while allowing demographic and built-environment associations to be evaluated separately and assessing whether the remaining associations are robust to the alternative specification. Heteroskedasticity-consistent HC3 standard errors were used for the OLS estimates [42].

Because the TRF characteristics exhibited significant spatial autocorrelation, spatial dependence in the regression residuals was subsequently evaluated using Global Moran's I. Significant residual spatial autocorrelation remained in the OLS specifications, indicating that the independence assumption of the cross-sectional errors was not satisfied [43]. We therefore estimated spatial error models (SEM) [38]:

$$Y_{ic} = a_c + \boldsymbol{X}_i'\boldsymbol{\beta}_c + u_{ic} \quad (16)$$

$$u_{ic} = \lambda_c \sum_j w_{ij}\, u_{jc} + \epsilon_{ic} \quad (17)$$

where u_{ic} denotes the spatially correlated error component, λ_c is the spatial error parameter, and w_{ij} represents the elements of the spatial-weight matrix. Consistent with the spatial autocorrelation analysis, spatial relationships were defined using a symmetrized four-nearest-neighbor matrix with row-standardized weights. Statistical significance of residual Moran's I was assessed using 999 random permutations. The SEM estimates were used as the primary basis for interpreting the regional correlates of TRF heterogeneity.

3. Results

3.1 Temperature response function

Municipality-specific temperature response functions exhibited predominantly U-shaped relationships between temperature and electricity demand for both residential and general-service consumers (Fig. 1). Among the 217 municipalities for which TRFs were successfully estimated, 210 residential TRFs (96.8%) and 196 general-service TRFs (90.3%) were classified as U-shaped over the common interpretation range of −10 to 32 °C. Thus, for most municipalities, electricity demand was lowest at an intermediate temperature and increased toward both lower and higher temperatures.

The location of the minimum response differed between the two contract types (Table 1). Among U-shaped TRFs, the mean optimal temperature was 16.6 °C for residential demand and 14.4 °C for general-service demand. The relative responses on the two sides of the optimum also differed between contract types. Residential demand exhibited a stronger heat-side than cold-side response, particularly toward the outer temperature range, whereas general-service demand showed more balanced cold- and heat-side sensitivities (Table 1). This asymmetric residential response is broadly consistent with Chang et al. [24], who also reported a U-shaped residential TRF for Korea with a substantially stronger demand response at high than at low temperatures.

Despite the predominance of U-shaped responses, substantial heterogeneity remained in the shapes of municipality-specific TRFs. The curves differed in their optimal temperatures, relative cold- and heat-side responses, and overall curvature. These cross-municipality differences motivated the functional decomposition of TRF shapes presented in the following section.

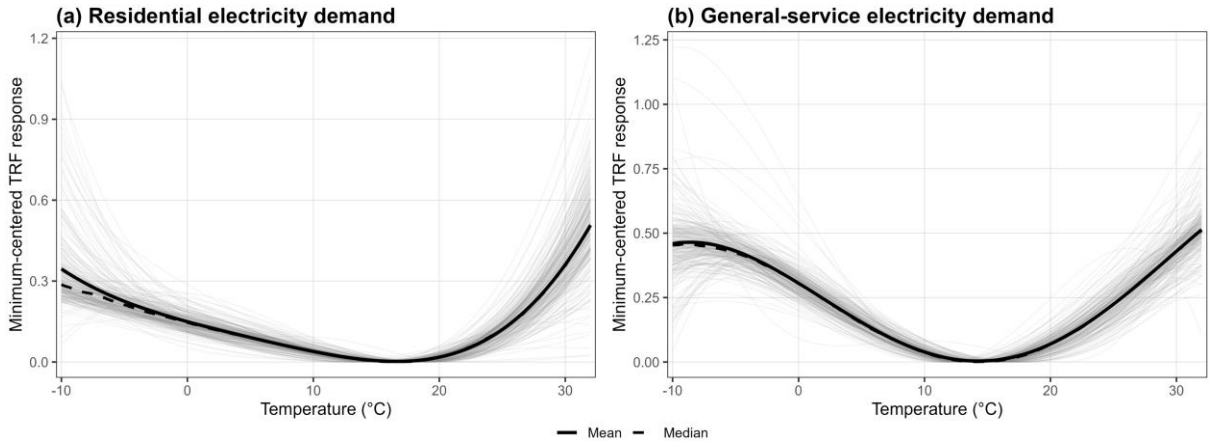


Fig. 1. Municipality-specific temperature response functions for (a) residential and (b) general-service electricity demand. Individual gray curves represent municipality-specific TRFs, while solid and dashed black lines indicate the cross-municipality mean and median, respectively. Each TRF is shifted vertically so that its minimum response within the displayed temperature range (−10 to 32 °C) equals zero; response amplitudes are otherwise preserved.

3.2 Functional variation in municipal TRFs

Functional principal component analysis was applied to the normalized U-shaped TRFs to characterize the dominant dimensions of cross-municipality variation in response shape. The analysis included 210 residential and 196 general-service TRFs. In both contract types, three principal components were required to exceed the 90% cumulative explained-variance threshold. The first three components jointly explained 95.8% of the variation in residential TRFs and 95.3% in general-service TRFs (Fig. 2), indicating that most of the cross-municipality variation in TRF shape could be represented in a low-dimensional functional space.

The distribution of explained variance differed between contract types. PC1–PC3 explained 62.4%, 26.5%, and 7.0% of residential variation, compared with 46.5%, 37.0%, and 11.8% for general-service demand, indicating a stronger concentration of residential variation in PC1.

The loading curves showed that residential PC1 primarily captured variation at low temperatures, whereas the remaining components reflected differences in relative cold- and heat-side responses and curvature. For general-service demand, the dominant components captured variation across both temperature extremes and contrasts between the cold and heat sides (Fig. 2). Because loading signs are arbitrary, interpretation focuses on their shapes and relative magnitudes.

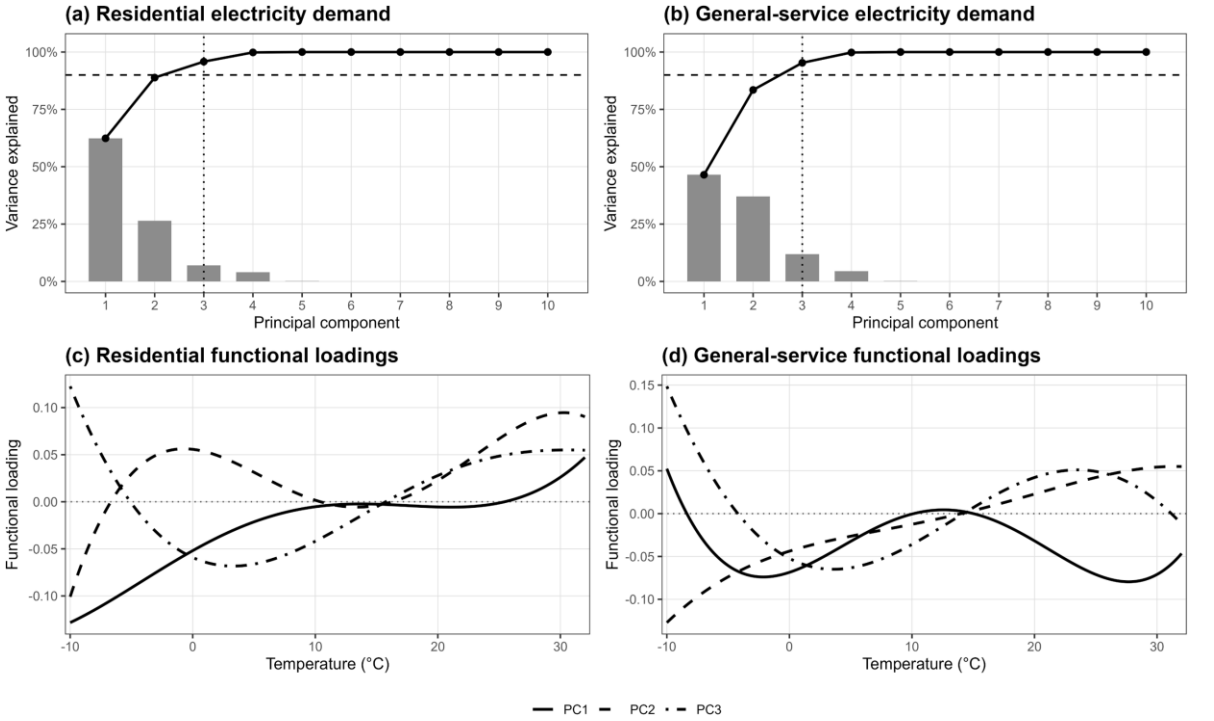


Fig. 2. Functional principal component analysis of municipality-specific temperature response functions. Panels (a) and (b) show the variance explained by individual principal components and the cumulative explained variance for residential and general-service demand, respectively. Panels (c) and (d) show the loading functions of the first three principal components. The horizontal dashed line indicates the 90% cumulative explained-variance threshold. Loading signs are arbitrary; interpretation therefore focuses on their shapes and relative magnitudes.

3.3 Functional clustering of municipal TRFs

The three-cluster solutions produced distinct normalized response profiles for both contract types, differing in optimal temperature and in the relative curvature of the cold- and heat-side responses (Fig. 3).

For residential demand, Cluster 1 exhibited comparatively moderate responses on both sides of the optimum and the smallest curve amplitude. Cluster 2 showed the strongest heat-side response and the largest amplitude, whereas Cluster 3 exhibited a relatively gradual cold-side response near the optimum but a substantially steeper response farther into the low-temperature range (Table 1, Fig. 3).

For general-service demand, Cluster 1 exhibited the strongest outer heat-side response and largest amplitude, whereas Cluster 2 was characterized by a stronger local cold-side response. Cluster 3 showed the strongest outer cold-side response. Notably, Clusters 1 and 2 had similar optimal temperatures but substantially different response profiles (Table 1, Fig. 3).

Overall, the clusters capture differences in complete TRF shape that cannot be summarized by optimal temperature alone, particularly in how responses steepen away from the optimum.

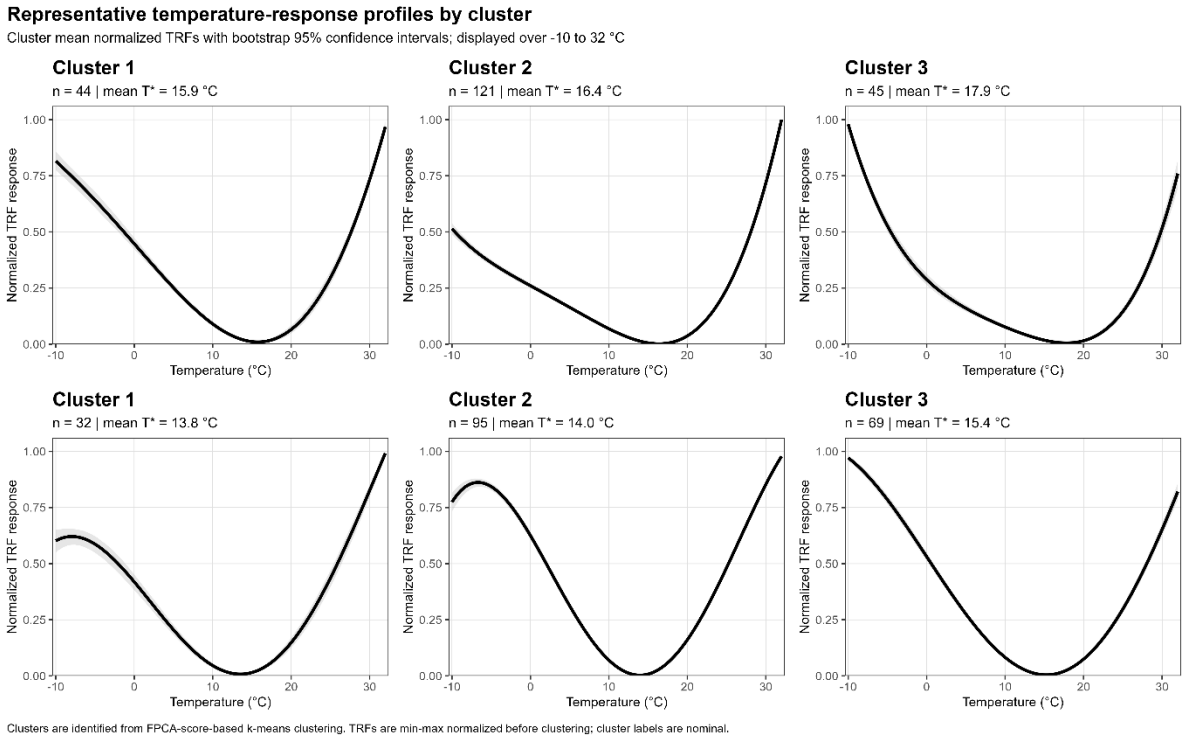


Fig. 3. Representative temperature-response profiles for the three TRF clusters. Panels show the mean min–max normalized TRF for each cluster of residential (top row) and general-service (bottom row) electricity demand. Shaded areas indicate pointwise 95% bootstrap confidence intervals. Cluster sample sizes and mean optimal temperatures (T^*) are reported above each panel. Cluster labels are nominal and do not imply an a priori ordering or interpretation.

Table 1
Characteristics of municipality-specific TRFs and functional clusters by electricity contract type.

Contract-cluster	n	Temperature-response characteristics						Climatic characteristics	
		Cold-side sensitivity		Heat-side sensitivity		Other characteristics			
		Outer cold	Local cold	Local heat	Outer heat	Optimal T (°C)	Amplitude	Mean T (°C)	Seasonal T SD (°C)
Res-All	210	0.0134 (0.0045)	0.0049 (0.0013)	0.0078 (0.0026)	0.0460 (0.0174)	16.6 (1.5)	0.534 (0.176)	12.7 (1.3)	9.7 (0.7)
Res-1	44	0.0125 (0.0017)	0.0048 (0.0011)	0.0063 (0.0019)	0.0305 (0.0120)	15.9 (1.8)	0.360 (0.092)	12.3 (1.5)	9.7 (0.7)
Res-2	121	0.0119 (0.0029)	0.0053 (0.0011)	0.0085 (0.0019)	0.0522 (0.0122)	16.4 (0.9)	0.595 (0.137)	12.7 (1.2)	9.9 (0.7)
Res-3	45	0.0182 (0.0063)	0.0039 (0.0016)	0.0075 (0.0040)	0.0445 (0.0233)	17.9 (1.7)	0.540 (0.218)	13.3 (1.3)	9.2 (0.6)
Gen-All	196	0.0238 (0.0065)	0.0099 (0.0023)	0.0106 (0.0022)	0.0373 (0.0106)	14.4 (1.4)	0.562 (0.116)	12.6 (1.2)	9.8 (0.7)
Gen-1	32	0.0201 (0.0040)	0.0096 (0.0015)	0.0112 (0.0026)	0.0449 (0.0166)	13.8 (1.9)	0.633 (0.148)	12.5 (1.2)	9.8 (0.8)

Gen-2	95	0.0225 (0.0049)	0.0108 (0.0016)	0.0112 (0.0015)	0.0344 (0.0052)	14.0 (0.8)	0.513 (0.057)	12.9 (1.0)	9.9 (0.6)
Gen-3	69	0.0272 (0.0078)	0.0087 (0.0029)	0.0096 (0.0023)	0.0378 (0.0109)	15.4 (1.3)	0.595 (0.132)	12.3 (1.5)	9.8 (0.7)

Note: Values are reported as mean (standard deviation). Local sensitivities are calculated within 5 °C of the optimal temperature (T^*), whereas outer sensitivities are calculated between the local range and the common endpoints (−8 °C and 32 °C). Kruskal–Wallis tests indicated significant differences among residential clusters in mean temperature ($H(2)=18.99$, $p<0.001$) and seasonal temperature variability ($H(2)=40.50$, $p<0.001$). For general-service clusters, mean temperature differed significantly ($H(2)=7.24$, $p=0.027$), whereas seasonal temperature variability did not ($H(2)=0.14$, $p=0.931$).

3.4 Spatial patterns of municipal TRFs

The spatial distribution of TRF clusters differed by contract type (Fig. 4). Residential Cluster 3 was relatively concentrated in southern municipalities, Cluster 1 occurred more frequently in central and northeastern regions, and Cluster 2 was broadly distributed nationwide. General-service clusters exhibited a more spatially mixed pattern.

Global Moran's I indicated significant positive spatial autocorrelation across all six residential TRF characteristics (Table 2), particularly for outer cold-side sensitivity and curve amplitude. Spatial autocorrelation was generally weaker for general-service demand, with optimal temperature and outer cold-side sensitivity exhibiting the strongest spatial structure. Overall, spatial organization was more pronounced for residential than for general-service demand.

Residential clusters differed significantly in both mean temperature and seasonal temperature variability, whereas general-service clusters differed only in mean temperature (Table 1).

Table 2
Spatial autocorrelation of TRF characteristics

TRF characteristics	Residential	General-service
Optimal temperature	0.356^{***}	0.320^{***}
Local cold-side sensitivity	0.375^{***}	0.150^{***}
Outer cold-side sensitivity	0.625^{***}	0.284^{***}
Local heat-side sensitivity	0.331^{***}	0.140^{***}
Outer heat-side sensitivity	0.398^{***}	0.178^{***}
Curve amplitude	0.459^{***}	0.080^{**}

Note: Entries report Global Moran's I. Statistical significance is based on 999 random permutations. *** p<0.01, **p<0.05, *p<0.10

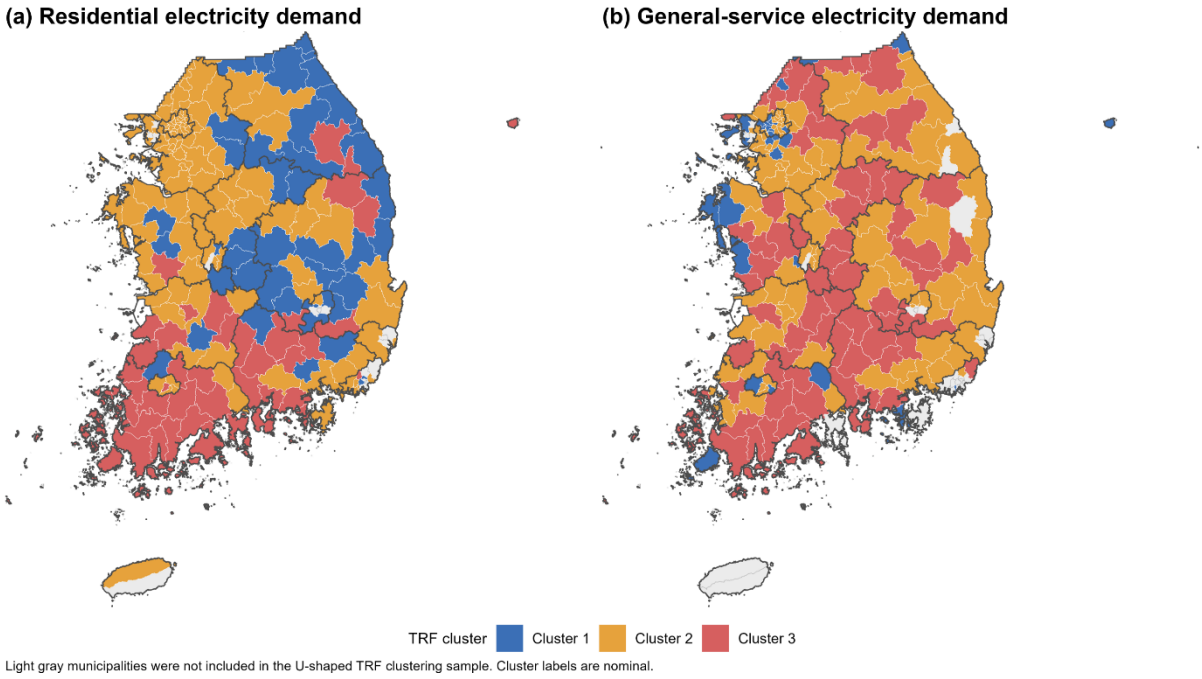


Fig. 4. Spatial distribution of municipality-specific TRF clusters for (a) residential and (b) general-service electricity demand. Colors indicate cluster membership obtained from FPCA-score-based k-means clustering of U-shaped TRFs. Light-gray municipalities were not included in the clustering sample. Cluster labels are nominal.

3.5 Regional correlates of TRF heterogeneity

Initial OLS models exhibited significant residual spatial autocorrelation across all specifications. In the SEM estimates, the spatial error parameter (λ) was positive and significant across all outcomes and specifications, while residual Moran's I was no longer significant, indicating that the SEM effectively captured the remaining spatial dependence (Table 3).

For residential demand, mean temperature was positively associated with optimal temperature in both specifications, while aging rate was also positively associated in Model A. Outer cold-side sensitivity was negatively associated with seasonal temperature variability; it was positively associated with aging rate in Model A and negatively associated with apartment share in Model B.

A clearer and more robust pattern emerged for outer heat-side sensitivity, which was negatively associated with seasonal temperature variability and positively associated with population density and real GRDP per capita in both specifications (Table 3).

For general-service demand, mean temperature was positively associated with optimal temperature, whereas population density was negatively associated with it in both specifications; apartment share was also negatively associated in Model B. Mean temperature was negatively associated with outer cold-side sensitivity, while seasonal temperature variability showed a weaker positive

association. Outer heat-side sensitivity was negatively associated with seasonal temperature variability in both specifications and with apartment share in Model B. Real GRDP per capita was not significantly associated with any of the three outcomes.

Overall, the correlates of TRF heterogeneity differed across response dimensions and contract types, with mean climatic conditions most consistently associated with the location of the demand minimum.

Table 3
Spatial error model estimates for regional correlates of municipality-specific TRF characteristics.

Contract	Model	Variable	Optimal temperature	Outer cold sensitivity	Outer heat sensitivity
Residential (N = 208)	Model A	Mean temperature	0.875*** (0.142)	0.0004 (0.0003)	-0.0017 (0.0016)
		Seasonal temperature SD	0.211 (0.132)	-0.0024*** (0.0004)	-0.0090*** (0.0016)
		Aging rate	0.375** (0.147)	0.0007** (0.0003)	0.0000
		Population density (log)	-0.188 (0.184)	0.0002 (0.0005)	0.0054** (0.0022)
		Real GRDP per capita (log)	-0.062 (0.089)	0.0001 (0.0002)	0.0027*** (0.0010)
		Spatial error parameter (λ)	0.326*** (0.093)	0.694*** (0.058)	0.550*** (0.074)
	Model B	Mean temperature	0.912*** (0.142)	0.0004 (0.0003)	-0.0019 (0.0016)
		Seasonal temperature SD	0.209 (0.134)	-0.0024*** (0.0004)	-0.0089*** (0.0016)
		Apartment share	-0.198 (0.123)	-0.0010*** (0.0003)	0.0012 (0.0013)
		Population density (log)	-0.365** (0.165)	0.0006 (0.0005)	0.0072*** (0.0020)
		Real GRDP per capita (log)	-0.102 (0.088)	0.0002 (0.0002)	0.0031*** (0.0010)
		Spatial error parameter (λ)	0.325*** (0.093)	0.710*** (0.056)	0.542*** (0.075)
General-service (N = 196)	Model A	Mean temperature	0.988*** (0.116)	-0.0029*** (0.0006)	0.0000 (0.0012)
		Seasonal temperature SD	0.210* (0.117)	0.0012* (0.0007)	-0.0038*** (0.0013)
		Aging rate	0.229* (0.120)	0.0008 (0.0007)	0.0009 (0.0013)
		Population density (log)	-0.942*** (0.151)	-0.0010 (0.0008)	0.0000 (0.0016)
		Real GRDP per capita (log)	-0.018 (0.075)	-0.0006 (0.0004)	0.0005 (0.0008)
		Spatial error parameter (λ)	0.270*** (0.099)	0.345*** (0.094)	0.324*** (0.095)
	Model B	Mean temperature	0.990*** (0.111)	-0.0028*** (0.0006)	-0.0001 (0.0012)
		Seasonal temperature SD	0.203* (0.114)	0.0012* (0.0007)	-0.0038*** (0.0012)
		Apartment share	-0.396*** (0.096)	-0.0006 (0.0005)	-0.0029*** (0.0010)
		Population density (log)	-0.823*** (0.127)	-0.0012 (0.0007)	0.0016 (0.0014)
		Real GRDP per capita (log)	0.001 (0.071)	-0.0006 (0.0004)	0.0007 (0.0008)
		Spatial error parameter (λ)	0.271*** (0.099)	0.349*** (0.094)	0.330*** (0.095)

Note: Entries report spatial error model coefficients, with standard errors in parentheses. All explanatory variables were standardized prior to estimation. Model A includes mean temperature, seasonal temperature variability, aging rate, log population density, and log real GRDP per capita. Model B replaces aging rate with apartment share. Spatial dependence was modeled using a symmetrized four-nearest-neighbor spatial-weight matrix with row-standardized weights. λ denotes the spatial error parameter. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.1$.

4. Conclusion and policy implications

This study examined the functional and spatial heterogeneity of temperature–electricity demand relationships across Korean municipalities by combining municipality-specific TRFs with functional and spatial analyses.

U-shaped responses predominated in both contract types, but substantial cross-municipality heterogeneity remained in optimal temperatures, sensitivities, and overall TRF shapes. Residential demand exhibited a particularly strong heat-side response, whereas general-service responses were more balanced. FPCA and clustering showed that this heterogeneity could be represented by a small number of recurring functional patterns that could not be characterized by optimal temperature alone.

TRF heterogeneity was also geographically structured, particularly for residential demand. Spatial regression further showed that mean temperature was positively associated with optimal temperature in both contract types, while stronger residential outer heat-side sensitivity was associated with lower seasonal temperature variability, higher population density, and higher real GRDP per capita.

These findings have several implications for electricity-system planning and climate adaptation. Most importantly, the substantial heterogeneity in municipality-specific TRFs cautions against applying a single national or highly aggregated temperature–demand relationship uniformly across regions. Even when municipalities have similar optimal temperatures, their responses to increasingly cold or hot conditions can differ substantially. Electricity-demand projections based solely on aggregate relationships may therefore understate demand responses in some municipalities while overstating them in others. Incorporating geographically differentiated TRFs into demand forecasting could provide a more realistic representation of regional weather sensitivity and improve assessments of future electricity requirements under changing temperature distributions.

The distinction between contract types is also relevant for planning. Strong residential heat-side responses suggest that more frequent high-temperature conditions may create substantial cooling pressures in some municipalities [9,17], while the spatial organization of cold-side responses indicates that winter electricity demand remains relevant despite a warming climate. The weaker spatial organization of general-service responses further cautions against applying a common spatial representation of weather sensitivity across consumer groups.

From a policy perspective, these findings support a more spatially differentiated approach to demand-side planning. Municipalities exhibiting strong heat-side responses may warrant greater attention to measures that reduce cooling-related peak demand, including building-energy-efficiency improvements, efficient cooling technologies, and demand-response programs during extreme heat. The spatial regression results further suggest that stronger residential heat-side sensitivity tends to occur in municipalities with higher population density and higher real GRDP

per capita, even after accounting for climatic conditions and residual spatial dependence. The positive association with population density is broadly consistent with previous evidence that urban overheating can increase cooling electricity consumption and peak power demand in built environments [39]. However, population density in the present analysis should not be interpreted as a direct measure of urban heat-island effects. Areas characterized by stronger cold-side responses may require continued consideration of winter peak management and the implications of electrified heating [11]. More broadly, the spatial clustering of similar TRF characteristics suggests that electricity-demand adaptation need not necessarily be designed solely at the national level; regional coordination among municipalities facing similar temperature-response profiles may provide a useful intermediate scale for demand forecasting, infrastructure planning, and climate-resilience strategies.

Several limitations should nevertheless be recognized. Although the spatial regression analysis identifies systematic associations between municipality-specific TRF characteristics and observable climatic, demographic, economic, and built-environment characteristics, these relationships are cross-sectional and should not be interpreted as causal effects. The significant spatial error parameters further indicate that unobserved geographically structured factors remain, potentially reflecting differences in building characteristics, heating and cooling technologies, appliance ownership, electricity-use patterns, or other local conditions not captured in the present analysis. The estimated relationships are also time-invariant and therefore do not capture potential changes in temperature sensitivity arising from adaptation, technological change, or evolving patterns of electricity use [8,11]. In particular, the positive association between mean climate and optimal temperature should not by itself be interpreted as evidence of climatic adaptation. Future research could extend the present framework by incorporating richer information on building stocks, heating and cooling technologies, appliance ownership, and other local characteristics, as well as by allowing temperature-response functions to evolve over time.

CRediT authorship contribution statement

Seungho Jeon: Conceptualization, Methodology, Data curation, visualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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